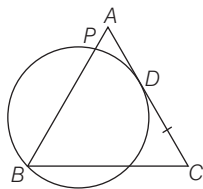


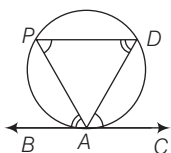
Sol. c. By using theorem,

$$\begin{aligned} AB \times AP &= AD^2 \\ &= \left(\frac{AC}{2}\right)^2 = \frac{1}{4}(AC)^2 \\ \Rightarrow AB \times AP &= \frac{1}{4}(AB)^2 \\ [\because AC &= AB \text{ given}] \\ \Rightarrow AB &= 4AP \end{aligned}$$

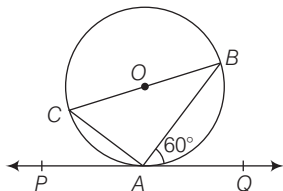


Theorem 14 Alternate Segment

Theorem In the given figure B, if BAC is a tangent to a circle at point A and if AD is any chord, then $\angle DAC = \angle APD$ and $\angle PAB = \angle PDA$.



EXAMPLE 7. In the given figure, PAQ is the tangent. BC is the diameter of the circle. If $\angle BAQ = 60^\circ$, then $\angle ABC$ is



- a. 25° b. 30° c. 45° d. 50°

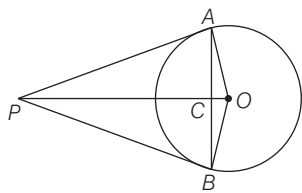
Sol. b. Since BC is the diameter of the circle.

$$\begin{aligned} \therefore \angle BAC &= 90^\circ & [\because \text{angle in a semi-circle is } 90^\circ] \\ \angle ACB &= 60^\circ & [\because \text{alternate angle theorem}] \end{aligned}$$

$$\begin{aligned} \text{Now, } \angle ACB + \angle BAC + \angle ABC &= 180^\circ \\ \Rightarrow \angle ABC + 90^\circ + 60^\circ &= 180^\circ \\ \Rightarrow \angle ABC &= 180^\circ - (90^\circ + 60^\circ) = 30^\circ \end{aligned}$$

- Theorem 15** If two tangents are drawn to a circle from an external point, then they subtend equal angles at the centre. These two tangents are equally inclined to the line segment joining the centre to that point. In the given figure,

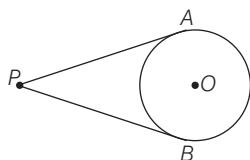
$$\begin{aligned} \angle AOP &= \angle BOP & \text{and} \\ \angle APO &= \angle BPO \end{aligned}$$



- Theorem 16** Line joining centre and external point is perpendicular bisector of line joining point of contacts of tangents to the circle. In the above figure, $OP \perp AB$ and $AC = BC$

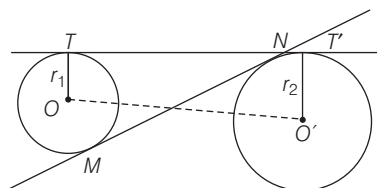
- Theorem 17** The length of two tangents drawn from an external point to a circle are equal.

$$\text{Here, } PA = PB$$



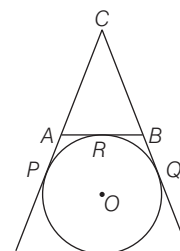
- Theorem 18** Tangents drawn at the end points of a diameter of a circle are always parallel.

- Length of the direct common tangent is $TT' = \sqrt{(OO')^2 - (r_1 - r_2)^2}$ and length of transverse common tangent is $MN = \sqrt{(OO')^2 - (r_1 + r_2)^2}$



EXAMPLE 8. In the given figure, CP and CQ are tangent to a circle with centre O . ARB is another tangent touching the circle at R . If $CP = 11$ cm, $BC = 7$ cm length of BR is

- a. 1 cm b. 2 cm
c. 4 cm d. 3 cm



Sol. c. Since, the length of two tangents drawn from external point are equal.

$$\therefore CQ = CP = 11 \text{ cm and } BQ = BR$$

$$\begin{aligned} \text{Now, } BQ &= CQ - BC \\ &= (11 - 7) \text{ cm} = 4 \text{ cm} \end{aligned}$$

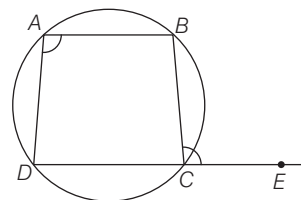
$$\therefore BR = BQ = 4 \text{ cm}$$

Cyclic Quadrilateral

A quadrilateral whose all vertices lie on a circle is called a cyclic quadrilateral.

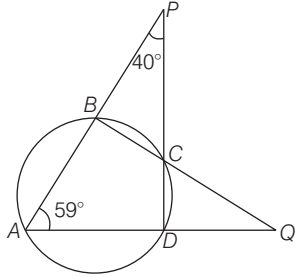
Properties of Cyclic Quadrilateral

- The sum of either pair of opposite angles of a cyclic quadrilateral is 180° i.e. the opposite angles of a cyclic quadrilateral are supplementary.
- If the sum of a pair of opposite angles of a quadrilateral is 180° , then quadrilateral is cyclic.
- The exterior angle, formed by producing a side of a cyclic quadrilateral is equal to the interior opposite angle.



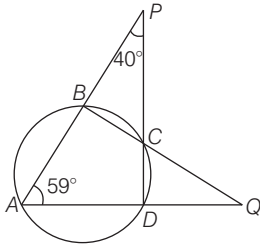
$$\text{Here, } \angle BAD = \angle BCE$$

EXAMPLE 9. In the given figure, if $\angle PAQ = 59^\circ$, $\angle APD = 40^\circ$, then what is $\angle AQB$?



- a. 19° b. 20° c. 22° d. 27°

Sol. c. Given, $\angle PAD = 59^\circ$ and $\angle APD = 40^\circ$



In $\triangle APD$, $\angle PAD + \angle APD + \angle ADP = 180^\circ$
 $\Rightarrow 59^\circ + 40^\circ + \angle ADP = 180^\circ \Rightarrow \angle ADC = 81^\circ$
 Also, $\angle ABC + \angle ADC = 180^\circ$
 [property of cyclic quadrilateral]
 $\Rightarrow \angle ABC = 180^\circ - 81^\circ = 99^\circ$
 $\therefore \angle CDQ = \angle ABC = 99^\circ$ and $\angle QCD = \angle BAD = 59^\circ$
 In $\triangle CQD$, $\angle CQD + \angle CDQ + \angle QCD = 180^\circ$
 $\Rightarrow \angle CQD = 180^\circ - 59^\circ - 99^\circ = 180^\circ - 158^\circ = 22^\circ$
 $\Rightarrow \angle AQB = \angle CQD = 22^\circ$

LOCUS

The path traced by a point moving under a given condition is called locus.

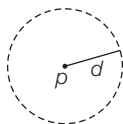
e.g. The locus of points (in a plane) equidistant from a fixed point (in a plane) is a circle with the fixed point as centre.

The plural of locus is loci.

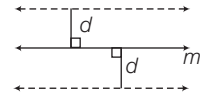
- The locus may be a line segment or lines, straight or curved.
- Every point which satisfies the given geometrical condition(s) lies on the locus and *vice-versa*.

Basic Locus Theorem

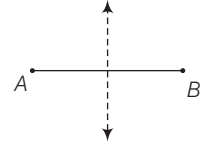
- The locus of points at a fixed distance d from point p is a circle with the given point p as its centre and d as its radius.



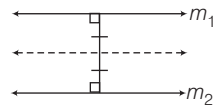
- The locus of points at a fixed distance d from a line m is a pair of parallel lines at a distance of d from line m and located on either side of m .



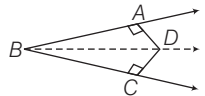
- The locus of points equidistant from two points A and B is the perpendicular bisector of the line segment joining the two points.



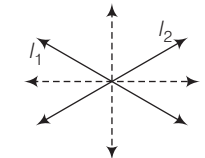
- The locus of points equidistant from two parallel lines m_1 and m_2 , is a line parallel to both m_1 and m_2 and midway between them.



- The locus of points in the interior of an angle equidistant from the sides of an angle is the bisector of the angle.

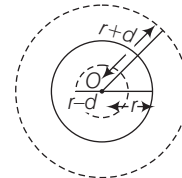


- The locus of points equidistant from two intersecting lines l_1 and l_2 is a pair of bisectors that bisect the angles formed by l_1 and l_2 .

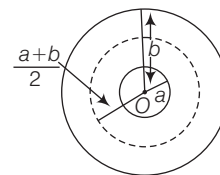


Locus in Some Standard Cases

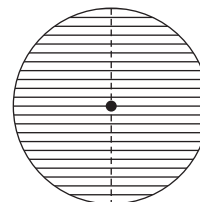
- The loci of all points d units from a circle having a radius r units are two concentric circles with radii of $r - d$ and $r + d$ units respectively.



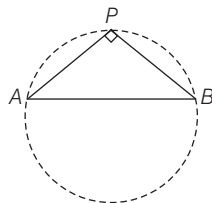
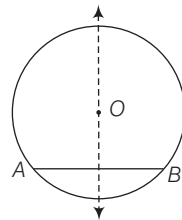
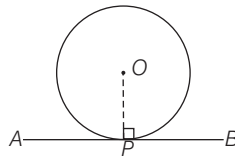
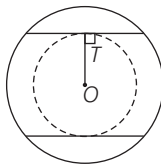
- The locus of all points equidistant from two concentric circles is a third circle whose radius is the average of the radii of the given circles.



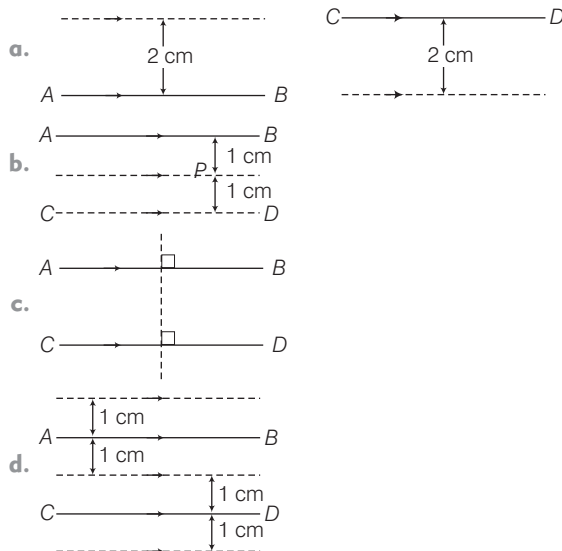
- The locus of the mid-points of all parallel chords of a circle is the diameter of the circle, which is perpendicular to the given chords.



4. The locus of the mid-points of equal chords is a circle with the same centre as the given circle and radius equal to the distance from the centre of the given circle to the given chord.
5. The locus of the centre of circle touching a given line at a given point is a line perpendicular to the given line through the given point.
6. The locus of the centre of all circles passing through two given points is the perpendicular bisector of the line segment joining the two points.
7. If A, B are fixed points, then the locus of a point P such that $\angle APB = 90^\circ$ is the circle with AB as diameter.



EXAMPLE 10. Given two parallel straight lines AB and CD which are 2 cm apart which of the following dotted lines may represent the locus of a point P such that it is equidistant from AB and CD ?

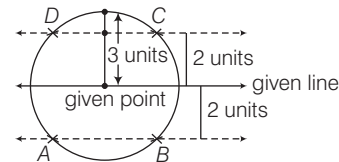


Sol. b.

EXAMPLE 11. What is the number of points in a plane two units from a given line and three units from a given point of the line?

- a. 1 b. 2 c. 3 d. 4

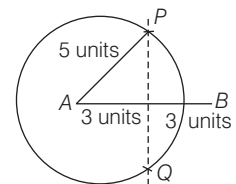
Sol. d. The answer will be 4 points. The dotted lines are first locus condition and the circle is the second locus condition. These two loci intersect at points marked with A, B, C and D .



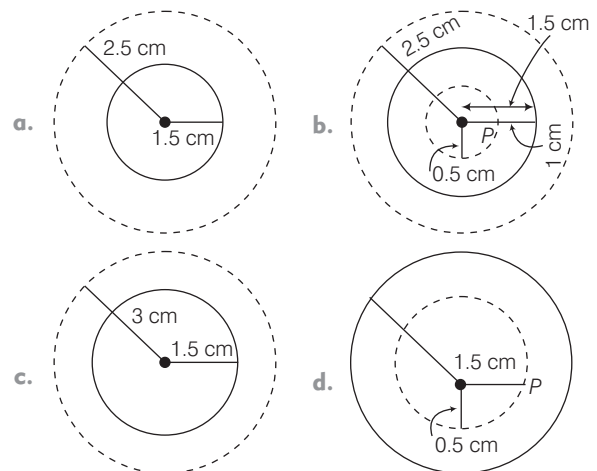
EXAMPLE 12. Two points A and B are 6 units apart. How many points are there that are equidistant from both A and B and also 5 units from A ?

- a. 0 b. 1 c. 2 d. 3

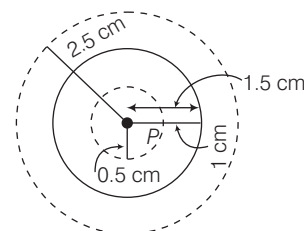
Sol. c. The answer will be 2 points. The dotted line is first locus condition and the circle is the second locus condition. These two loci intersect at two points marked with P and Q .



EXAMPLE 13. Given a circle with radius 1.5 cm. Which of the following dotted lines may represent the locus of a point P such that P is at a distance of 1 cm from the nearest point on the circle?



Sol. b.

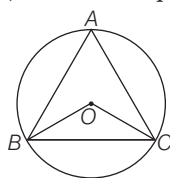


PRACTICE EXERCISE

1. In a circle with centre O and radius 5 cm, AB is a chord of length 8 cm. If $OM \perp AB$, then what is the length of OM ?

(a) 4 cm (b) 5 cm
(c) 3 cm (d) None of these

2. An equilateral $\triangle ABC$ is inscribed in a circle with centre O . Then, $\angle BOC$ is equal to



(a) 120° (b) 75° (c) 180° (d) 60°

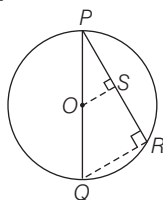
3. A square $ABCD$ is inscribed in a circle with centre O . Then, the angle subtended by each side of the square at the centre O is

(a) 120° (b) 180° (c) 45° (d) 90°

4. The points of concurrency of the perpendicular bisectors of the sides of a triangle is called its

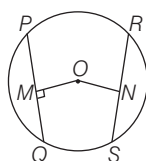
(a) incentre (b) circumcentre
(c) excentre (d) orthocentre

5. In the given figure, PQ is the diameter of a circle with centre at O . OS is perpendicular to PR . Then, OS is equal to



(a) $\frac{1}{4}QR$ (b) $\frac{1}{3}QR$ (c) $\frac{1}{2}QR$ (d) QR

6. In the given figure, OM and ON are the perpendiculars drawn on the chords PQ and RS . If $OM = ON = 6$ cm. Then,



(a) $PQ \geq RS$ (b) $PQ < RS$ (c) $PQ \leq RS$ (d) $PQ = RS$

7. The locus of the mid-points of all radii of a circle is a

(a) circle (b) parallelogram
(c) rhombus (d) square

8. The perpendicular bisectors of the sides of a triangle pass through the

(a) different point (b) more than 2 points
(c) same point (d) None of these

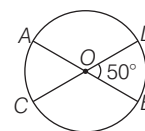
9. If a point P moves such that the sum of the squares of its distance from two fixed points A and B is a constant, then the locus of P is

(a) a circle
(b) a straight line
(c) an arbitrary curve
(d) the perpendicular bisector of AB

10. The locus of points equidistant from two fixed points is a straight line which

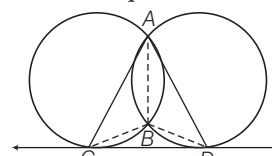
(a) is at right angles to the line joining the two fixed points
(b) bisects the line joining the two fixed points
(c) is the perpendicular bisector of the line joining the two fixed points
(d) None of the above

11. Diameter AB and CD of a circle intersect at O . If $m\angle BOD = 50^\circ$, then $m\angle AOD$ is



(a) 50° (b) 180° (c) 130° (d) 310°

12. CD is a direct common tangent to two circles intersecting each other at A and B . Then, $\angle CAD + \angle CBD$ is equal to?



(a) 180° (b) 90° (c) 360° (d) 120°

13. In a circle of radius 17 cm, two parallel chords are drawn on opposite sides of a diameter. The distance between the chords is 23 cm. If the length of one chord is 16 cm, then the length of the other chord is

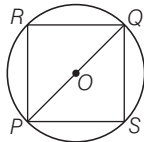
(a) 34 cm (b) 15 cm (c) 23 cm (d) 30 cm

14. If AB is a chord of a circle, P and Q are the two points on the circle different from A and B , then

(a) the angles subtended at P and Q by AB are always equal
(b) the sum of the angles subtended by AB at P and Q is always equal to two right angles
(c) the angles subtended by AB at P and Q are either equal or supplementary
(d) None of the above

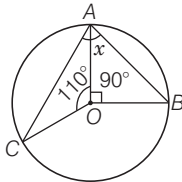
- 15.** The locus of the centre of circles which passes through two given points is
 (a) perpendicular to the line joining the given points at one of those points
 (b) perpendicular bisector of the line joining the given points
 (c) parallel to the line joining the given points
 (d) None of the above

- 16.** In the adjoining figure, POQ is the diameter of the circle, R and S are any two points on the circles. Then,



- (a) $\angle PRQ > \angle PSQ$ (b) $\angle PRQ < \angle PSQ$
 (c) $\angle PRQ = \angle PSQ$ (d) $\angle PRQ = \frac{1}{2} \angle PSQ$

- 17.** If O is the centre of the circle, then the value of x in the adjoining figure is

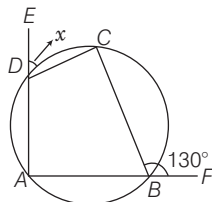


- (a) 80° (b) 70° (c) 60° (d) 50°

- 18.** If S is a circle with centre C and P be a movable point outside S , then the locus of P such that the tangents from P to S are of constant length is

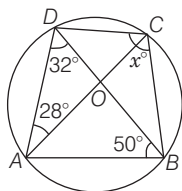
- (a) the straight line CP
 (b) the circle through P with centre at C
 (c) a circle intersecting S
 (d) a circle touching S

- 19.** In the given figure A, B, C, D are the concyclic points. The value of x is



- (a) 50° (b) 60° (c) 70° (d) 90°

- 20.** If O is the centre of the circle, then x is



- (a) 72° (b) 62° (c) 82° (d) 52°

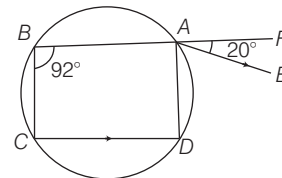
- 21.** Two circles touch each other internally. Their radii are 2 cm and 3 cm. The biggest chord of the outer circle which is outside the inner circle is of length

- (a) $2\sqrt{2}$ cm (b) $3\sqrt{2}$ cm (c) $2\sqrt{3}$ cm (d) $4\sqrt{2}$ cm

- 22.** If two circles are such that the centre of one lies on the circumference of the other, then the ratio of the common chord of the two circles to the radius of any one of the circles is

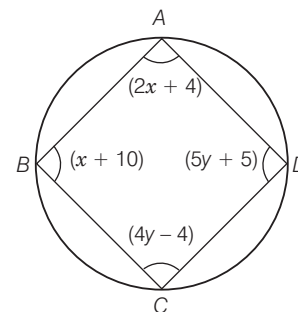
- (a) $2 : 1$ (b) $\sqrt{3} : 1$ (c) $\sqrt{5} : 1$ (d) $4 : 1$

- 23.** In the given figure, $ABCD$ is a cyclic quadrilateral. AE is drawn parallel to CD and BA is produced. If $\angle ABC = 92^\circ$ and $\angle FAE = 20^\circ$, then $\angle BCD$ is equal to



- (a) 88° (b) 98° (c) 108° (d) 72°

- 24.** The values of $x + y$ in the figure is equal to

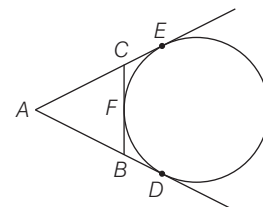


- (a) 90° (b) 85° (c) 75° (d) 65°

- 25.** Let A and B be two points. What is the locus of the point P such that $\angle APB = 90^\circ$?

- (a) The line AB itself
 (b) The point P itself
 (c) The circumference of the circle with AB as diameter
 (d) The line perpendicular to AB and bisecting AB

- 26.** In the adjoining figure AD, AE and BC are tangent to the circle at D, E, F respectively, then



- (a) $AD = AB + BC + AC$ (b) $2AD = AB + BC + AC$
 (c) $AD = \frac{1}{4}(AB + BC + AC)$ (d) $3AD = AB + BC + AC$